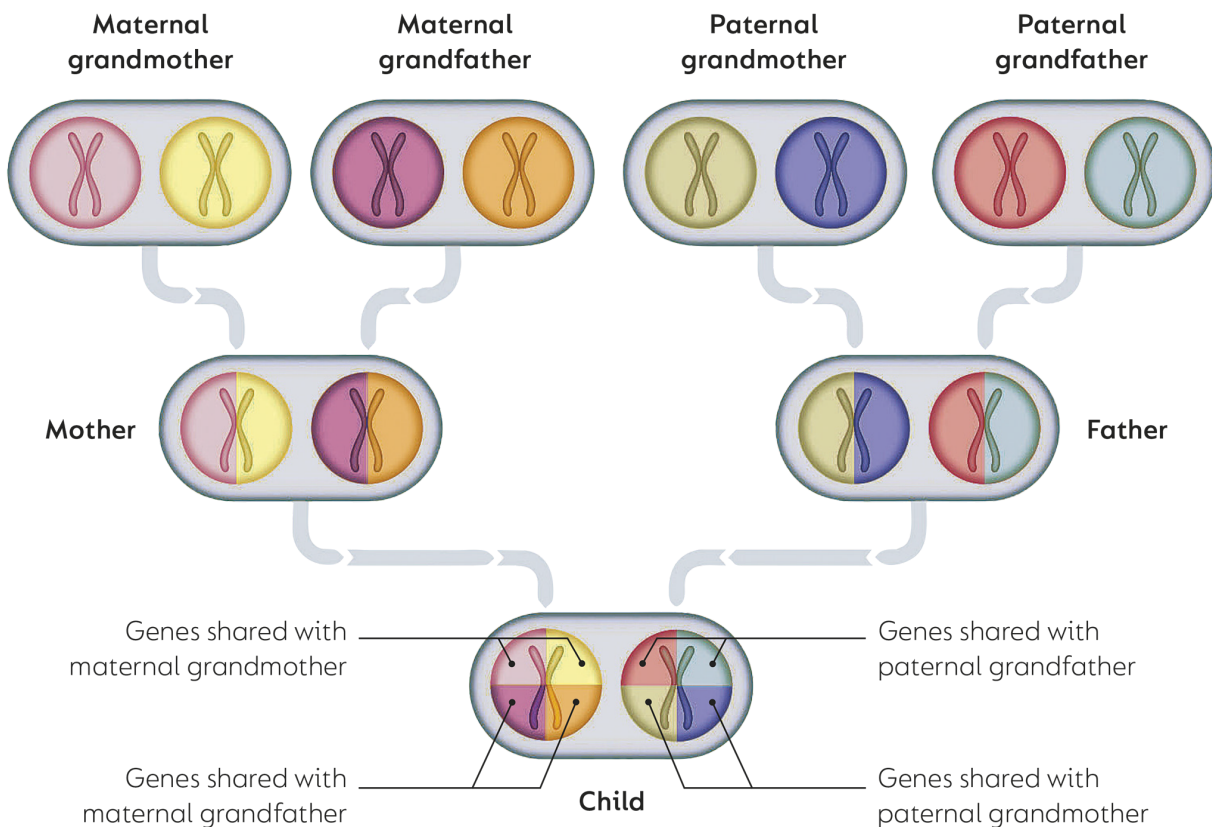


GENES THROUGH THE GENERATIONS

Through each generation, genes are shuffled and reshuffled. Half of a baby's genes come from its father and half from its mother. The baby's parents in turn inherited half each of their genes from each of their own parents. One quarter of each person's genes therefore come from the grandparents. So how does this happen? Instead of containing the full complement of 46 chromosomes, each egg and each sperm has just half, or 23, chromosomes each. When they meet at conception the chromosomes pair up to again make up the full complement that now forms the genetic blueprint for the new individual. One of the 23 pairs of chromosomes are sex chromosomes, so sex is also determined at conception. Each egg carries an X chromosome and each sperm either an X or Y. If two X chromosomes combine, the baby will be a girl; if an X and Y chromosome combine, the baby will be a boy (see below).



DOMINANT AND RECESSIVE GENES

Genes come in pairs, and each gene in a pair may differ slightly. One gene may be dominant, and override the other gene, which is recessive. A recessive gene only has an effect if both genes in a pair are recessive. An example is eye color, although this is not as simple as depicted here.

KEY



Recessive gene for blue eyes

Dominant gene for brown eyes